Set 40: Naming and drawing hydrocarbons

Notes

The name of any simple organic compound is based on its structure. It ensures that given the structure, a name can be constructed and given the name a structure can be drawn. The rules that govern this naming convention were adopted and first published by the Union of Pure and Applied Chemistry (IUPAC) in 1958. This set of rules has become known as the IUPAC system of nomenclature for organic compounds. The rules used here include those revised in 1993.

Aliphatic hydrocarbons

For straight chain (aliphatic) hydrocarbons the rules can be summarised as follows.

1. The prefix of the name is used to identify the number of carbon atoms in a continuous chain, that is, the chain length.

Number of carbon atoms	Prefix	Prefix for alkyl group
1	meth-	methyl
2	eth-	ethyl
3	prop-	propyl
4	but-	butyl
5	pent-	pentyl
6	hex-	hexyl
7	hept-	heptyl
8	oct-	octyl

2. The suffix of the name is used to identify the type of aliphatic hydrocarbon, that is, alkane, alkene or alkyne.

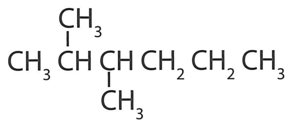
Type of hydrocarbon	Functional group	Suffix
alkane	$C-C$	-ane
alkene	$C=C$	-ene
alkyne	$C\equiv C$	-yne

3. Groups other than alkyl groups can result from addition and substitution reactions. The names of halogen groups are as follows.

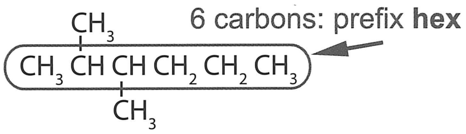
Type of substituted hydrocarbon	Functional group	Name of group
fluorocarbon	-F	fluoro-
chlorocarbon	-Cℓ	chloro-
bromocarbon	-Br	bromo-
iodocarbon	-I	iodo-

The steps for naming hydrocarbons

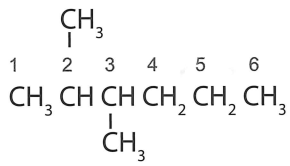
The following structure will be used to illustrate the rules for naming hydrocarbons in each step:



Step 1: Identify the longest continuous carbon chain containing the functional group. This determines the prefix for the name.



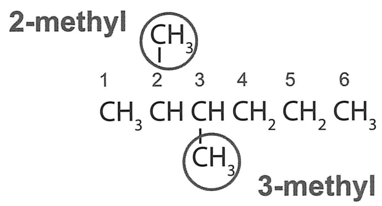
Step 2: Assign a number to each carbon atom in the longest continuous chain starting at the end of the chain that results in the lowest possible number for the position of the functional group.



Step 3: Identify the presence and location of any double or triple bond. This determines the suffix of the name: -ane (only single bonds), -ene (double bond), -yne (triple bond)

suffix: **- ane**
(only single bonds present)

Step 4: Identify all groups attached to the longest carbon chain.

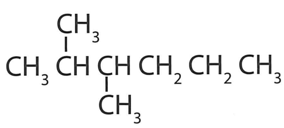


Step 5: Add the names of the groups in front of the chain name, indicating the position of the group using the number of the carbon to which it is attached. If a group occurs more than once, use the prefix di-, tri-, tetra-, etc to indicate the number present and indicate the position of each as described above. A number for the location of groups is often used even though it may be redundant. The locating number is placed as close to the part of the name referring to the group, as in butan-1-ene.

2,3-dimethyl

Step 6: Substituent groups are written alphabetically on the basis of the group name. Numbers are separated from each other by commas and numbers are separated from names by hyphens.

2,3-dimethylhexane



Set 40: Naming and drawing hydrocarbons

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The steps for naming hydrocarbons

The following compound will be used to illustrate the rules for drawing structural formulas of hydrocarbons: 1-fluoro-2-methylbutane

Step 1: Identify and draw a carbon skeleton of the parent chain.

1-fluoro-2-methyl butane

C - C - C - C

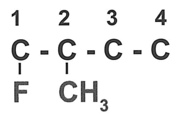
Step 2: Number the carbons in the chain.

1 2 3 4

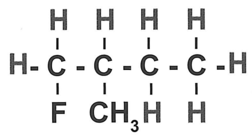
C - C - C - C

Step 3: Add the funtional groups to the appropriate carbons on the chain.

1-fluoro 2-methyl butane

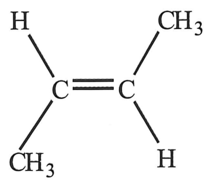


Step 4: Add hydrogen atoms to give each carbon four (4) bonds.

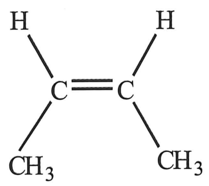


Geometric isomers

In alkenes, rotation about the double bond is not possible. This results in the existance of geometric isomers in which the atoms and their bonding is the same but the arrangement of atoms in space can be different. The two different isomers are distinguished by placing the prefix ***trans-*** or ***cis-*** in front of the name, as follows



trans-but-2-ene



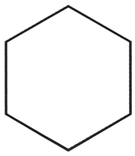
cis-but-2-ene

Alicyclic compounds

These are ring structures formed when the ends of a carbon chain join.
The rules for naming alicyclic hydrocarbons are mostly the same as for those used to name aliphatic hydrocarbons. There are some differences though, which include the following.

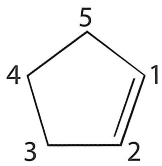
1. The prefix cyclo- is placed in front of the name indicating the number of carbon atoms.

   cyclohexane (C_6H_{12})



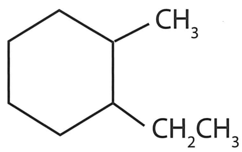
2. The carbon atoms are numbered starting at the functional group and for double and triple bonds the numbering is such that the carbon atoms either side of the multiple bond have the numbers 1 and 2.

   cyclopentene (C_5H_8)



3. If more than one type of group is attached the numbering starts at the group that is first in alphabetical order.

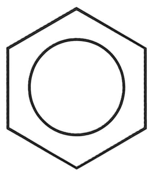
   1-ethyl-2-methylcyclohexane (C_9H_{18})



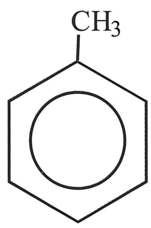
Aromatic compounds based on benzene

The naming of these compounds is very similar to the alicyclic compounds and because the compounds we will consider are based on the benzene structure, the names are based on the name benzene. For example:

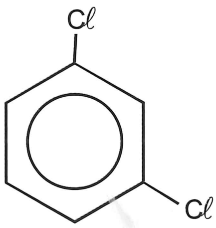
(a) benzene (C_6H_6)



(b) methylbenzene, also known as toluene (C_7H_8)



(c) 1,3-dichlorobenzene ($C_6H_4C\ell_2$)



Notes

Set 40: Naming and drawing hydrocarbons

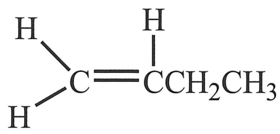
Notes

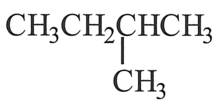
Set 40: Exercises

Oil is a complex mixture of hydrocarbons and can include some of the substances in **questions 1 and 2**.

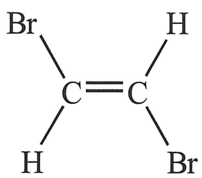
1. Write systematic names for the following compounds.

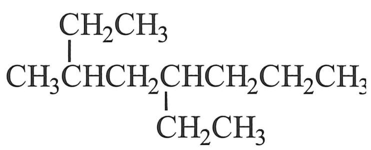
(a) $CH_3CH_2CH_2CH_2CH_3$

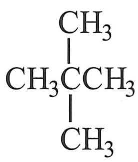
(b) 

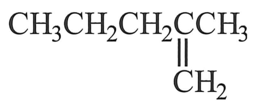
(c) 

(d) $CHC\ell_2CH_2C\ell$

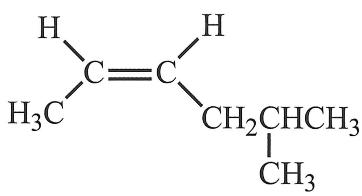
(e) 

(f) 

(g) 

(h) 

(i) $CF_3CH_2CH_2CH_2CH_3$

(j) 

2. Draw structural formulae for the following compounds.
   (a) 2,2,4-trimethylpentane
   (b) dichlorodifluoromethane
   (c) 3-ethyl-2-methylpent-2-ene
   (d) 4,4-diethyloctane
   (e) 5,5-dichloro-4-methylhex-1-ene
   (f) *trans*-hept-3-ene
   (g) 1,1-dichloro-*cis*-but-2-ene
   (h) 5-ethylhept-1-ene

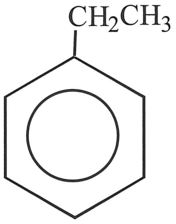
3. Oil contains many substances that have the same formula but different structures. Draw the structural isomers and write systematic names for
   (a) all the isomers of:
   (i) pentane
   (ii) pentene
   (b) four isomers of C_4H_9Br

Alicyclic and aromatic compounds are often used as starting materials for important agricultural chemicals including pesticides and herbicides. Examples of substituted alicyclic and aromatic compounds are included in **questions 4 and 5**.

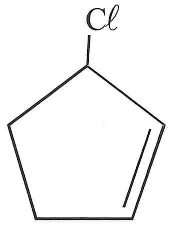
Notes

4. Write systematic names for the following compounds.

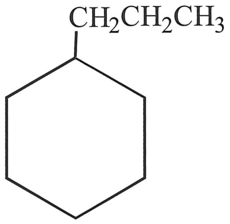
(a)



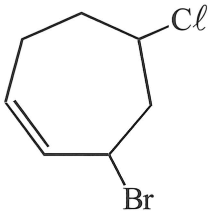
(b)



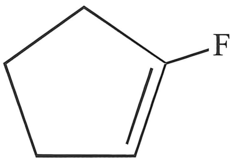
(c)



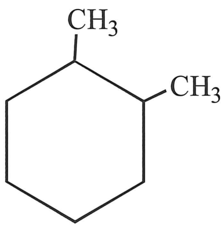
(d)



(e)



(f)



5. Draw structural formulae for the following compounds.
   - (a) fluorocycloheptane
   - (b) 4-methylcyclopentene
   - (c) butylbenzene
   - (d) 1,2-difluorobenzene
   - (e) 1,3-dibromobenzene
   - (f) 1-ethyl-4-methylbenzene

6. (a) Oil contains a number of different compounds with the formula C_4H_8
   Draw all the isomers of C_4H_8

   (b) Benzene with some of its hydrogen atoms substituted with halogens can be a starting material for the synthesis of pesticides.
   Draw all the isomers of dichlorobenzene.